

1. Administrative & Study Identification

Study Title: Sacituzumab Tirumotecan (MK-2870) as Monotherapy and in Combination With Pembrolizumab in Participants With Advanced Solid Tumors (MK-2870-008) **Official Title:** Phase 1 Trial of MK-2870 as Monotherapy and in Combination With Pembrolizumab ± Chemotherapy in Subjects With Advanced Solid Tumors **Short Title/ Acronym:** TroFuse-008 (MK-2870-008) **Protocol Number:** MK-2870-008 / jRCT2031230356 **NCT Number:** NCT06049212 **Posting ID:** 309937517547042970 **Sponsor / Company:** Merck (Merck Sharp & Dohme LLC) **Investigational Product:** Sacituzumab tirumotecan (MK-2870, TROP2-directed ADC) ± pembrolizumab (drug preferred name: pembrolizumab; Trade Name: Keytruda; ATC Code: L01FF02; Preferred UMLS Name: Immune Checkpoint Inhibitors; EMA URL: https://www.ema.europa.eu/en/documents/product-information/keytruda-epar-product-information_en.pdf) **Phase of Development:** PHASE1 **Trial Status:** RECRUITING (Currently recruiting participants) **Medical Monitor:** Merck Sharp & Dohme LLC Clinical Director

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, and dose-limiting toxicities (DLTs) of sacituzumab tirumotecan (MK-2870) as monotherapy and in combination with pembrolizumab in participants with advanced solid tumors and advanced/metastatic non-small cell lung cancer (NSCLC). **Secondary Objectives:** To assess the Objective Response Rate (ORR), Duration of Response (DOR), Progression-Free Survival (PFS), Overall Survival (OS), and specific Pharmacokinetic (PK) metrics of sacituzumab tirumotecan.

2.2 Background & Rationale To address the need for effective targeted therapies in patients with previously treated advanced solid tumors and treatment-naïve advanced/metastatic NSCLC. The study aims to evaluate if combining a TROP2-directed antibody-drug conjugate (sacituzumab tirumotecan) with an immune checkpoint inhibitor (pembrolizumab) can safely enhance anti-tumor activity and improve clinical outcomes for these patient populations.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Phase ID / Criteria Type:** PHASE1 / PHASE_REQUIREMENT (Criteria Text: Trial must be in PHASE1) **Status Name / Description:** RECRUITING / Currently recruiting participants **Control Type:** Single-arm

(Multi-cohort safety/expansion study) **Blinding:** Open-label
Randomization: N/A (Parallel assignment to distinct cohorts/arms)

3.2 Planned Enrollment & Duration Targeted Enrollment: 30 participants
Number of Sites: Multicenter (focused on Japanese participants)
Estimated Study Start: 26-Oct-2023
Estimated Primary Completion: 30-Jun-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically or cytologically confirmed advanced/metastatic solid tumor by pathology report (Arm 1) or advanced/metastatic NSCLC (Arm 2; Stage IIIB/IIIC not candidates for surgical resection/chemoradiation, or Stage IV). 2. For Arm 1: Have received, or been intolerant to, all treatment known to confer clinical benefit. AEs due to previous therapies must have recovered to $\leq$ Grade 1 or baseline. 3. For Arm 2: Confirmation that EGFR-, ALK-, or ROS1-directed therapy is not indicated as primary therapy. 4. For Arm 2: Tumor tissue demonstrates PD-L1 TPS $\geq 50\%$ as determined by PD-L1 IHC 22C3 pharmDx assay by local laboratory. 5. Measurable disease by RECIST 1.1 as assessed by the local site investigator/radiology. 6. Archival tumor tissue sample or newly obtained core/incisional biopsy provided. 7. Have a life expectancy of at least 3 months and an ECOG performance status of 0 or 1. 8. Agree to highly effective contraception and refrain from sperm donation during the intervention period and required clearance times (100 days for sacituzumab tirumotecan, 90 days for carboplatin); females must not be pregnant or breastfeeding.

4.2 Exclusion Criteria 1. Known active CNS metastases, carcinomatous meningitis, or symptomatic ascites/pleural effusion (clinically stable participants are eligible). 2. History of (noninfectious) pneumonitis/interstitial lung disease that required steroids, or active pneumonitis/ILD. 3. Active autoimmune disease requiring systemic treatment in the past 2 years, or active infection requiring systemic therapy. 4. Grade ≥ 2 peripheral neuropathy. 5. Severe dry eye syndrome, severe Meibomian gland disease, or severe corneal disease. 6. Active or past inflammatory bowel disease requiring immunosuppressive medication. 7. Uncontrolled cardiovascular disease (e.g., NYHA Class III/IV heart failure, QTcF >480 ms) within 6 months. 8. Prior treatment with a TROP2-targeted ADC or topoisomerase I-containing ADC. 9. Arm 1: Discontinued prior immunotherapy due to $\geq$ Grade 3 irAE, Grade 2 myocarditis, or recurrent Grade 2 pneumonitis. 10. Arm 2: Prior therapy with anti-PD-1, anti-PD-L1, anti-PD-L2, or other T-cell targeted agents, or prior systemic chemotherapy/targeted therapy for metastatic NSCLC. 11. Positive for HIV, concurrent active Hepatitis B, or Hepatitis C. 12. Required treatment with strong CYP3A4 inhibitors/inducers within 14 days, or live/live-attenuated vaccine within 30 days before the first dose.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Arm 1:** Sacituzumab tirumotecan (MK-2870) administered once every 2 weeks (Q2W). * **Arm 2:** Sacituzumab tirumotecan Q2W combined with pembrolizumab once every 6 weeks (Q6W). * **Arm 3 Note:** Per protocol amendment 04, Arm 3 (Pembrolizumab/Carboplatin + sacituzumab tirumotecan Combination Therapy) was discontinued, and all related procedures and recruitment were removed. **Route of Administration:** Intravenous (IV) Infusion **Duration of Treatment:** Continued until disease progression, unacceptable toxicity, or patient withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for oncology patients. **Prohibited:** Strong inhibitors or inducers of CYP3A4, live or live-attenuated vaccines, and concurrent investigational agents or systemic immunosuppressive therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Imaging, Pathology Review, Safety Labs
Baseline	Day 0	First Dose (MK-2870 + Pembrolizumab), PK sampling, Safety Baseline
Interim	Q2W / Q6W	Onsite Visits: IV Dosing, Safety Labs, PK/ADA collection, Tumor Assessments
End of Study	Up to 100 days	Final Safety Follow-up, Survival tracking, Exit Interview

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. **Number of Participants Experiencing Dose-Limiting Toxicity (DLT):** Graded via NCI-CTCAE v5.0 during the DLT evaluation period. 2. **Number of Participants Experiencing ≥ 1 Adverse Event (AE):** Any untoward medical occurrence temporally associated with study intervention. 3. **Number of Participants Discontinuing from Study Therapy due to AE.**

7.2 Secondary & Exploratory Outcomes 1. **Area Under the Plasma Concentration-Time Curve (AUC)** of sacituzumab tirumotecan-ADC. 2. **Maximum Plasma Concentration (Cmax)** of sacituzumab tirumotecan-ADC. 3. **Minimum Plasma Concentration (Cmin)** of sacituzumab tirumotecan-ADC. (Additional exploratory metrics include ORR, PFS, DOR, and OS).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via onsite visits, graded per CTCAE v5.0 metrics. **Serious Adverse Events (SAEs):** Reported within 24 hours to the sponsor; safety profiles thoroughly tracked (including ILD/pneumonitis and neutropenia monitoring). **Data Monitoring Committee (DMC):** Regular safety review via an internal Safety Review Committee (SRC) designed to evaluate DLTs for dose clearance.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients who receive ≥ 1 dose) for primary DLT/AE evaluations. Efficacy Set (participants with measurable baseline disease) for ORR/PFS metrics. **Sample Size Justification:** Targeted enrollment of 30 subjects, powered adequately for an exploratory Phase 1 safety, tolerability, and PK signal-finding trial. **Handling of Missing Data:** Safety data summarized as observed; efficacy parameters (PFS, DOR) will be censored at the date of the last evaluable tumor assessment.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Rigorous onsite and remote source data verification (SDV) by the Sponsor's clinical monitoring team. **Audit Trail:** Standardized electronic Data Capture (eDC) system utilization, with eCRF locking and tracked data clarification forms.

11. Study Identifiers & Governance

Primary ID: MK-2870-008 **Registry Number:** NCT06049212 **Posting ID:** 309937517547042970 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** TroFuse-008 / MK-2870-008

12. Clinical Framework (Oncology Specific)

Disease Areas / Conditions: Non-small Cell Lung Cancer, Solid Tumors, Programmed Cell Death-1 (PD1, PD-1), Programmed Cell Death 1 Ligand 1(PDL1, PD-L1), Programmed Cell Death 1 Ligand 2 (PDL2, PD-L2). **Disease Name / Preferred Name:** Non-Small Cell Lung Carcinoma **Definition:** A heterogeneous aggregate of at least three distinct histological types of lung cancer, including SQUAMOUS CELL CARCINOMA; ADENOCARCINOMA; and LARGE CELL CARCINOMA. They are dealt with collectively because of their shared treatment

strategy. **Semantic Type:** Neoplastic Process **Diagnosis Threshold:** Advanced/Metastatic (Stage IIIB, IIIC, or IV) **Medical Methodology:** TROP2-directed Antibody-Drug Conjugate (ADC) combined with PD-1 Checkpoint Inhibition. **Terminology Codes & Hierarchy (Source Level: 5):** * **MeSH Code:** D002289 * **HPO Code:** HP: 0030358 * **SNOMED CT ID:** 254637007 * **SNOMED Hierarchy:** 448993007| Carcinoma of lung|Malignant epithelial neoplasm of lung (disorder) \$\rightarrow\$ 39607008|Lung structure|Lung structure (body structure) \$\rightarrow\$ 128632008|Non-small cell carcinoma|Non-small cell carcinoma (morphologic abnormality) \$\rightarrow\$ 64572001|Disease|Disease (disorder)

13. Study Procedures & Methodology

Management Pathway: Phase 1 Oncology Dose Escalation & Expansion Pathway **Education Protocol (HCP):** Site initiation visits and investigator meetings focused on study drug administration and immune-mediated AE management **Education Protocol (Patient):** Informed consent process, dosing calendars, and proactive AE reporting guidelines **Quality Audit Mechanism:** Clinical monitoring, DLT tracking, and continuous data review by the Safety Review Committee

14. Observation & Follow-Up Schedule

Total Duration: Estimated completion January 2027 (including follow-up) **Onsite Visit Cadence:** Every 2 weeks (Q2W) to coincide with MK-2870 infusions **Remote Monitoring Frequency:** Continuous safety oversight mapped to clinical schedules **Checklist Review Interval:** Tumor imaging assessment sweeps iteratively performed per RECIST 1.1 intervals (usually Q8W)

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				